

TheMITbro GATE 2027 Electrical Engineering — Set 01

Set 01 Question Paper — Human-QA Review Copy

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Paper ID: GATE_2027_EE_SET_01

Manifest content SHA-256: 0a5a9c14cdca6c92 db41317239a36afe 73da6efda2b5ae64
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Questions / marks / duration: 65 / 100 / 180 minutes

Instructions

- Questions 1-10 are General Aptitude; Questions 11-65 are the selected subject.
- Each question carries either 1 mark or 2 marks, as printed beside it.
- For MCQ, select exactly one option. A wrong 1-mark MCQ attracts 1/3 negative mark; a wrong 2-mark MCQ attracts 2/3 negative mark.
- MSQ and NAT questions have no negative marking. This project reference specifies no partial marks for MSQ.
- For NAT, enter only the numerical answer in the permitted answer box on the examination interface.
- Internal audit IDs appear only because this is a review copy; they must be removed from the commercial learner copy.

Section A — General Aptitude (10 questions, 15 marks)

Q1. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-GA-002 · revision 1 · Medium

In a vocabulary-in-context review, the committee described its approval as “qualified” because two safety tests were still unresolved. Here, the word “qualified” most nearly means

- **A.** unconditional
 - **B.** limited by conditions
 - **C.** professionally trained
 - **D.** officially certified
-

Q2. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-GA-003 · revision 1 · Medium

Read the following passage and answer the reading-comprehension question. An office tested a flexible starting-time policy for four weeks. Among the 120 employees who voluntarily submitted travel data, the recorded mean commute duration fell from 52 minutes before the pilot to 46 minutes during it. No travel data were collected from nonparticipants. Which conclusion is best supported?

- **A.** The policy reduced the commute duration of every employee.
 - **B.** Among the reporting employees, the recorded mean commute duration was 6 minutes lower during the pilot.
 - **C.** The nonparticipants experienced the same mean reduction as the participants.
 - **D.** The pilot proves that traffic volume across the entire city decreased.
-

Q3. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-GA-004 · revision 1 · Hard

Read the following passage and select all supported statements. A fault-prediction model scored 68% on a fixed held-out sample from Plant B. After recalibration using a separate training sample from Plant B, it scored 81% on that same held-out sample. No post-recalibration test was performed at Plant A.

- **A.** The measured accuracy on the held-out Plant B sample increased by 13 percentage points.
 - **B.** Recalibration guarantees improved accuracy on every future case from Plant B.
 - **C.** The post-recalibration accuracy at Plant A is unknown from the given information.
 - **D.** Relative to the original 68% score, the measured increase is approximately 19.1%.
-

Q4. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-GA-005 · revision 1 · Medium

For a narrative-sequencing task, arrange the following sentences into the most coherent order. P: Only then could results from repeated tests be compared. Q: The initial instruction merely asked the team to improve reliability. R: The team therefore defined failure as more than two interruptions per 100 operating hours. S: Without a numerical criterion, however, success could not be judged.

- **A.** Q-S-R-P
 - **B.** Q-R-S-P
 - **C.** S-Q-P-R
 - **D.** R-Q-S-P
-

Q5. [1 mark; NAT]

Audit ID: TMB-GATE-EE-GA-008 · revision 1 · Medium

A data-interpretation table gives inspection results for three quarters.

Quarter	Accepted	Rejected
Q1	84	16
Q2	108	12
Q3	126	14

Enter the overall acceptance percentage, rounded to two decimal places.

NAT response: _____**Q6. [2 marks; MCQ]**

Audit ID: TMB-GATE-EE-GA-010 · revision 1 · Medium

In a permutations-and-combinations problem, six distinct jobs A, B, C, D, E and F are arranged in a sequence. Job A must occur before job B, and jobs C and D must not be adjacent. The number of valid sequences is

- **A.** 120
- **B.** 180
- **C.** 240
- **D.** 360

Q7. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-GA-013 · revision 1 · Medium

For a logic-deduction task, consider the premises: (i) every relay trip is logged; (ii) some logged events are manually generated; and (iii) no manually generated event is a fault. Select all conclusions that must follow.

- **A.** No manually generated event is a fault.
- **B.** Some logged events are not faults.
- **C.** Some relay trips are not faults.
- **D.** Every logged event is a relay trip.

Q8. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-GA-014 · revision 1 · Medium

In an analytical-aptitude scheduling problem, four inspections P, Q, R and S occupy positions 1 through 4, one per position. P is before Q, R is immediately after P, and S is not fourth. Which statement must be true?

- **A.** P is first.
- **B.** Q is fourth.
- **C.** R is second.
- **D.** S is first.

Q9. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-GA-016 · revision 1 · Hard

For a logic-deduction task, use these premises: (i) every device that passed test T was calibrated; (ii) some calibrated devices failed test S; and (iii) no device that failed test S was shipped. Select all conclusions that necessarily follow.

- **A.** Some calibrated devices were not shipped.
 - **B.** Every shipped device passed test T.
 - **C.** Some devices that passed test T were shipped.
 - **D.** No uncalibrated device passed test T.
-

Q10. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-GA-019 · revision 1 · Medium

In a three-dimensional spatial-aptitude problem, a cube has distinct face labels P, Q, R, S, T and U. The opposite face pairs are P-U, Q-T and R-S. Select all triples that can meet at a single vertex.

- **A.** P, Q, R
 - **B.** P, U, T
 - **C.** Q, R, S
 - **D.** P, T, S
-

Section B — Engineering Mathematics (8 questions, 13 marks)**Q11. [1 mark; MCQ]**

Audit ID: TMB-GATE-EE-EM-001 · revision 2 · Easy

Determine the largest eigenvalue of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

- **A.** 1
 - **B.** 2
 - **C.** 3
 - **D.** 4
-

Q12. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-EM-003 · revision 2 · Hard

Let P be a real symmetric idempotent matrix, i.e. $P^T = P$ and $P^2 = P$. Which of the following statements are necessarily true?

- **A.** Every eigenvalue of P is either 0 or 1.
 - **B.** P is positive semidefinite.
 - **C.** $\text{tr}(P) = \text{rank}(P)$.
 - **D.** $\det(P) = 1$.
-

Q13. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-EM-004 · revision 2 · Easy

Evaluate the exponential limit $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$.

- **A.** 1
 - **B.** 2
 - **C.** 4
 - **D.** Does not exist
-

Q14. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-EM-008 · revision 2 · Medium

Consider the second-order linear ordinary differential equation $y'' + 4y' + 3y = 0$. Which of the following functions are solutions?

- **A.** e^{-x}
 - **B.** e^{-3x}
 - **C.** e^{-2x}
 - **D.** $2e^{-x} - 5e^{-3x}$
-

Q15. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-EM-009 · revision 2 · Easy

Evaluate $(1 + i)^4$.

- **A.** 4
 - **B.** -4
 - **C.** $4i$
 - **D.** $-4i$
-

Q16. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-EM-013 · revision 2 · Medium

Let a random variable X have finite mean μ and variance σ^2 . Which statements are necessarily true?

- **A.** $E[X - \mu] = 0$
 - **B.** $E[X^2] = \sigma^2 + \mu^2$
 - **C.** $\text{Var}(aX + b) = a^2\sigma^2$ for constants a, b
 - **D.** $E[(X - \mu)^2] = \sigma^2$
-

Q17. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-EM-015 · revision 3 · Hard

Let C be the positively oriented contour $|z| = 2$. Evaluate $\oint_C \frac{z^2 + 1}{z(z - 1)} dz$.

- A. 0
 - B. $2\pi i$
 - C. $4\pi i$
 - D. $-2\pi i$
-

Q18. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-EM-018 · revision 2 · Hard

For the planar vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$, which statements are true?

- A. $\nabla \cdot \mathbf{F} = 0$
 - B. $\nabla \times \mathbf{F} = -2\mathbf{k}$
 - C. $\mathbf{F}$ is conservative on $\mathbb{R}^2$
 - D. The counter-clockwise circulation along $x^2 + y^2 = 1$ is -2π .
-

Section C — Electrical Engineering (47 questions, 72 marks)**Q19. [1 mark; NAT]**

Audit ID: TMB-GATE-EE-NT-001 · revision 2 · Easy

A $6\ \mu\text{F}$ capacitor and a $3\ \mu\text{F}$ capacitor are connected in series across an 18 V DC source. Determine the total energy stored by this capacitor network; enter the numerical value in mJ.

NAT response: _____

Q20. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-NT-002 · revision 2 · Medium

Two coupled inductors in a series-aiding network have $L_1 = 4\ \text{H}$, $L_2 = 9\ \text{H}$ and coefficient of coupling $k = 0.5$. Determine the equivalent inductance.

- A. 7 H
 - B. 13 H
 - C. 19 H
 - D. 25 H
-

Q21. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-NT-003 · revision 2 · Hard

At the input port of a linear network, a $4\ \Omega$ resistor is connected from the port to ground. A dependent current source of value $0.5i_R$, where i_R is the resistor current, is also directed from the port to ground. Determine the resistance seen at the port.

- A. $2\ \Omega$
 - B. $\frac{8}{3}\ \Omega$
 - C. $4\ \Omega$
 - D. $6\ \Omega$
-

Q22. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-NT-004 · revision 2 · Easy

Select every correct statement about ideal passive network elements under finite, impulse-free excitation; one or more options may be correct.

- A. The energy stored in a capacitor is $\frac{1}{2}Cv^2$.
 - B. The energy stored in an inductor is Li^2 .
 - C. A capacitor voltage cannot change instantaneously.
 - D. An inductor current cannot change instantaneously.
-

Q23. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-005 · revision 2 · Medium

In a resistive network, node v_1 is connected to a 12 V source through $3\ \Omega$, to ground through $6\ \Omega$, and to node v_2 through $2\ \Omega$. Node v_2 is connected to ground through $4\ \Omega$. Using KCL and node-voltage analysis, determine v_2 ; enter the numerical value in V.

NAT response: _____**Q24. [2 marks; NAT]**

Audit ID: TMB-GATE-EE-NT-006 · revision 2 · Hard

Two non-reference node voltages satisfy $v_1 - v_2 = 5\text{ V}$ because of an ideal voltage source between them. A $2\ \Omega$ branch connects v_1 to ground, a $3\ \Omega$ branch connects v_2 to ground, and a 4 A current source injects current into v_2 . Using supernode KCL, determine v_1 ; enter the numerical value in V.

NAT response: _____**Q25. [2 marks; MCQ]**

Audit ID: TMB-GATE-EE-NT-007 · revision 2 · Medium

Clockwise mesh currents I_1 and I_2 in a resistor network satisfy $5I_1 - 2I_2 = 8$ and $-2I_1 + 4I_2 = 2$. Determine the current in their common branch, directed with I_1 .

- A. 0.625 A
 - B. 1.625 A
 - C. 2.250 A
 - D. 3.875 A
-

Q26. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-NT-008 · revision 2 · Easy

Select every correct statement about node-voltage analysis of a connected electrical network; one or more options may be correct.

- **A.** One node is selected as the reference node.
 - **B.** An ideal voltage source between two non-reference nodes can be handled using a supernode.
 - **C.** For n nodes, at most $n - 1$ independent KCL node equations are required.
 - **D.** The method cannot be used when dependent sources are present.
-

Q27. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-NT-009 · revision 2 · Medium

An 18 V ideal source feeds a $3\ \Omega$ series resistor and an output node; a $6\ \Omega$ resistor connects that node to ground. A $4\ \Omega$ load is then connected from the output node to ground. Determine the load current using the Thevenin equivalent of the source network.

- **A.** 1 A
 - **B.** 2 A
 - **C.** 3 A
 - **D.** 4 A
-

Q28. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-NT-010 · revision 2 · Easy

A Norton network consists of a 3 A current source in parallel with $6\ \Omega$. When a $3\ \Omega$ load is connected across the network, determine the load current.

- **A.** 0.5 A
 - **B.** 1 A
 - **C.** 2 A
 - **D.** 3 A
-

Q29. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-011 · revision 2 · Hard

A sinusoidal network has Thevenin RMS voltage $V_{\text{th}} = 10\text{ V}$ and Thevenin impedance $Z_{\text{th}} = (3 + j4)\ \Omega$. A conjugately matched load is connected. Determine the maximum average load power; enter the numerical value in W.

NAT response: _____

Q30. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-NT-012 · revision 2 · Medium

Select every correct statement about superposition in a linear electrical network; one or more options may be correct.

- **A.** An independent ideal voltage source is replaced by a short circuit when deactivated.
 - **B.** An independent ideal current source is replaced by an open circuit when deactivated.
 - **C.** Dependent sources remain active.
 - **D.** The total power cannot in general be obtained by adding the powers caused by the individual sources.
-

Q31. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-013 · revision 2 · Medium

In a first-order RC network, the capacitor voltage is initially 2 V and its steady-state value after switching is 10 V. The resistance is $2\text{ k}\Omega$ and the capacitance is $100\text{ }\mu\text{F}$. Determine the capacitor voltage one time constant after switching; enter the numerical value in V.

NAT response: _____

Q32. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-014 · revision 2 · Medium

A 12 V DC source is applied at $t = 0$ to an initially unenergized series RL network with resistance $R = 4\text{ }\Omega$ and inductance $L = 2\text{ H}$. Determine the inductor current at $t = (0.5 \ln 2)\text{ s}$; enter the numerical value in A.

NAT response: _____

Q33. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-NT-015 · revision 2 · Easy

A series RLC network has inductance $L = 50\text{ mH}$ and capacitance $C = 20\text{ }\mu\text{F}$. Determine its undamped resonant angular frequency.

- **A.** 100 rad s^{-1}
 - **B.** 500 rad s^{-1}
 - **C.** 1000 rad s^{-1}
 - **D.** 5000 rad s^{-1}
-

Q34. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-NT-016 · revision 2 · Medium

A series RLC network has resistance $R = 20\text{ }\Omega$ and inductance $L = 0.1\text{ H}$. Determine its half-power bandwidth in angular frequency.

- **A.** 100 rad s^{-1}
 - **B.** 200 rad s^{-1}
 - **C.** 500 rad s^{-1}
 - **D.** 2000 rad s^{-1}
-

Q35. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-NT-017 · revision 2 · Hard

A single-phase impedance load absorbs $P = 8 \text{ kW}$ at a lagging power factor of 0.8. Select every correct statement about this sinusoidal steady-state network; one or more options may be correct.

- **A.** The load reactive power is 6 kVAR.
 - **B.** The apparent-power magnitude is 10 kVA.
 - **C.** The load current leads the load voltage.
 - **D.** A 6 kVAR shunt capacitor would correct the source power factor to unity.
-

Q36. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-018 · revision 2 · Medium

A balanced three-phase, star-connected load has per-phase impedance $Z_{\text{ph}} = (8 + j6) \Omega$ and is supplied at a line-to-line RMS voltage $V_L = 400 \text{ V}$. Determine the total real power in kilowatts.

NAT response: _____

Q37. [2 marks; NAT]

Audit ID: TMB-GATE-EE-NT-019 · revision 2 · Hard

A two-port network has $z_{11} = 4 \Omega$, $z_{12} = z_{21} = 2 \Omega$, and $z_{22} = 5 \Omega$. Port 2 is terminated by a load $Z_L = 3 \Omega$. Determine the input impedance; enter the numerical value in Ω .

NAT response: _____

Q38. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-NT-020 · revision 2 · Medium

A balanced three-phase, delta-connected load has branch-impedance magnitude $|Z_{\Delta}| = 12 \Omega$ and lagging phase angle 30° . It is supplied at a line-to-line RMS voltage $V_L = 240 \text{ V}$. Determine the approximate total real power.

- **A.** 7.20 kW
 - **B.** 12.47 kW
 - **C.** 14.40 kW
 - **D.** 24.94 kW
-

Q39. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-SS-001 · revision 1 · Easy

The continuous-time signal $x(t) = \cos(4t) + \sin(6t)$ is periodic. Determine its fundamental period.

- **A.** $\frac{\pi}{3} \text{ s}$
 - **B.** $\frac{\pi}{2} \text{ s}$
 - **C.** $\pi \text{ s}$
 - **D.** $2\pi \text{ s}$
-

Q40. [1 mark; NAT]

Audit ID: TMB-GATE-EE-SS-002 · revision 1 · Easy

For the discrete-time signal $x[n] = (\frac{1}{2})^n u[n]$, determine the signal energy $\sum_{n=-\infty}^{\infty} |x[n]|^2$.**NAT response:** _____**Q41. [2 marks; MSQ]**

Audit ID: TMB-GATE-EE-SS-003 · revision 1 · Hard

A continuous-time system is defined by $y(t) = t x(t)$. Select all properties that this system satisfies.

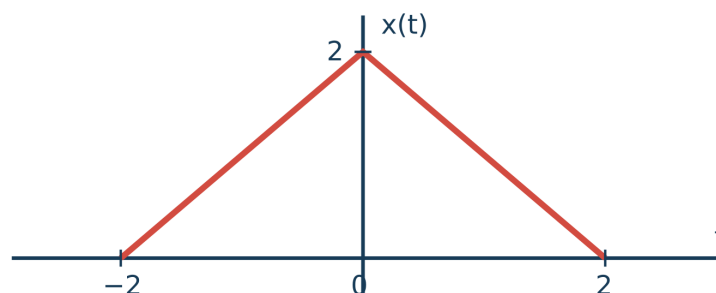
- **A.** Linear
- **B.** Time invariant
- **C.** Memoryless
- **D.** Causal

Q42. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-SS-004 · revision 1 · Medium

The triangular signal shown in the figure is $x(t) = 2 - |t|$ for $|t| \leq 2$ and zero otherwise. Determine its energy.

- **A.** $\frac{8}{3}$
- **B.** $\frac{4}{3}$
- **C.** $\frac{16}{3}$
- **D.** $\frac{8}{3}$

Figure: Triangular signal $x(t)$

Accessible description: Triangular signal with vertices at (-2,0), (0,2), and (2,0).

Q43. [2 marks; NAT]

Audit ID: TMB-GATE-EE-SS-005 · revision 1 · Medium

Let $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$. If $y(t) = x(t) * h(t)$, determine $y(\ln 2)$.**NAT response:** _____

Q44. [1 mark; NAT]

Audit ID: TMB-GATE-EE-SS-006 · revision 1 · Easy

A real LTI system has frequency response $H(j2) = 3e^{j\pi/6}$. For input $x(t) = \cos(2t + \pi/18)$, enter the amplitude of the steady-state output sinusoid.

NAT response: _____

Q45. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-SS-007 · revision 1 · Medium

For a discrete-time LTI system, $x[n] = h[n] = u[n] - u[n - 3]$. Determine the output sample $y[2]$.

- **A.** 1
 - **B.** 2
 - **C.** 3
 - **D.** 4
-

Q46. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-SS-008 · revision 1 · Medium

The periodic waveform shown in the figure equals $+2$ for $0 \leq t < T/4$ and -2 for $T/4 \leq t < T$. Select all correct statements.

- **A.** Its average value is -1 .
- **B.** Its RMS value is 2.
- **C.** Its average power is 4.
- **D.** Its DC component is zero.

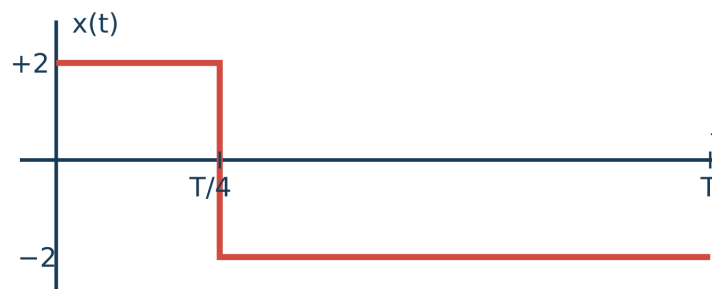


Figure: Two-level periodic waveform

Accessible description: One period of a waveform at $+2$ for the first quarter-period and -2 for the remaining three quarters.

Q47. [2 marks; NAT]

Audit ID: TMB-GATE-EE-SS-009 · revision 1 · Hard

The periodic input to a real LTI system is $x(t) = 2 + 4 \cos(\omega_0 t) + 3 \sin(2\omega_0 t)$. The frequency-response magnitudes are $|H(0)| = \frac{1}{2}$, $|H(j\omega_0)| = 1$, and $|H(j2\omega_0)| = 2$. Determine the average output power.

NAT response: _____

Q48. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-SS-010 · revision 1 · Easy

The signal $x(t) = e^{-2t}u(t) - e^{3t}u(-t)$ has a bilateral Laplace transform. Which region of convergence applies?

- **A.** $\text{Re}(s) > 3$
 - **B.** $\text{Re}(s) < -2$
 - **C.** $-2 < \text{Re}(s) < 3$
 - **D.** $\text{Re}(s) > -2$
-

Q49. [2 marks; NAT]

Audit ID: TMB-GATE-EE-SS-011 · revision 1 · Medium

A causal signal has Laplace transform $X(s) = \frac{s+4}{s(s+2)(s+3)}$. Determine its final value $\lim_{t \rightarrow \infty} x(t)$.

NAT response: _____

Q50. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-SS-012 · revision 1 · Medium

For $X(z) = \frac{z}{z-0.5} - \frac{z}{z-2}$ with region of convergence $|z| > 2$, determine $x[n]$.

- **A.** $[(0.5)^n - 2^n]u[n]$
 - **B.** $[(0.5)^n - 2^n]u[-n-1]$
 - **C.** $(0.5)^n u[n] + 2^n u[-n-1]$
 - **D.** $-(0.5)^n u[-n-1] - 2^n u[n]$
-

Q51. [2 marks; MSQ]

Audit ID: TMB-GATE-EE-SS-013 · revision 1 · Hard

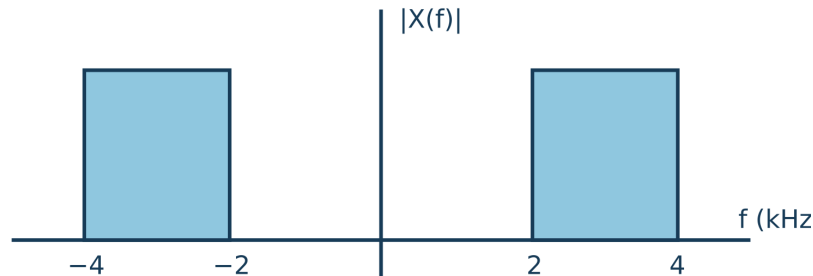
Consider $H(z) = \frac{1-0.5z^{-1}}{(1-0.25z^{-1})(1-2z^{-1})}$. Select all correct statements about possible regions of convergence.

- **A.** For $|z| > 2$, the system is causal but unstable.
 - **B.** For $0.25 < |z| < 2$, the system is stable and two sided.
 - **C.** For $|z| < 0.25$, the system is anti-causal and unstable.
 - **D.** The system is FIR for every possible region of convergence.
-

Q52. [1 mark; NAT]

Audit ID: TMB-GATE-EE-SS-014 · revision 1 · Medium

The magnitude spectrum shown in the figure is nonzero only over $2 \leq |f| \leq 4$ kHz. Using the conventional baseband sampling theorem, enter the minimum sampling frequency in kHz that guarantees no aliasing.

NAT response: _____Figure: Magnitude spectrum $|X(f)|$

Accessible description: Two rectangular spectral bands from -4 to -2 kHz and from 2 to 4 kHz.

Q53. [2 marks; MCQ]

Audit ID: TMB-GATE-EE-SS-015 · revision 1 · Hard

The signal $x(t) = \cos(2\pi 1000t) + \cos(2\pi 2600t)$ is sampled at 4 kHz. The samples are passed through an ideal reconstruction low-pass filter with passband $0 \leq f \leq 2$ kHz. Which two positive-frequency components appear in the reconstructed signal?

- **A.** 0.6 kHz and 1.0 kHz
- **B.** 1.0 kHz and 1.4 kHz
- **C.** 1.0 kHz and 2.0 kHz
- **D.** 1.4 kHz and 2.6 kHz

Q54. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-EMFT-001 · revision 1 · Easy

A point charge Q is located at the centre of a closed spherical surface. Which value equals $\oint_S \mathbf{D} \cdot d\mathbf{S}$?

- **A.** 0
- **B.** $Q/(4\pi)$
- **C.** Q
- **D.** Q/ϵ_0

Q55. [1 mark; NAT]

Audit ID: TMB-GATE-EE-EMFT-002 · revision 1 · Easy

An infinite line charge in free space has uniform density 55.6 nC/m . Determine the electric-field magnitude at radial distance 0.1 m ; enter the value in kV/m . Use $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

NAT response: _____**Q56. [2 marks; MSQ]**

Audit ID: TMB-GATE-EE-EMFT-003 · revision 1 · Hard

The figure shows a charge-free boundary between dielectric 1 with $\epsilon_{r1} = 2$ and dielectric 2 with $\epsilon_{r2} = 5$. In medium 1, the electric field has normal component 3 V/m and tangential component 4 V/m . Select all correct statements for medium 2.

- **A.** $D_{2n} = D_{1n}$
- **B.** $E_{2t} = E_{1t}$
- **C.** $E_{2n} = 1.2 \text{ V/m}$
- **D.** $|\mathbf{E}_2| = 5 \text{ V/m}$

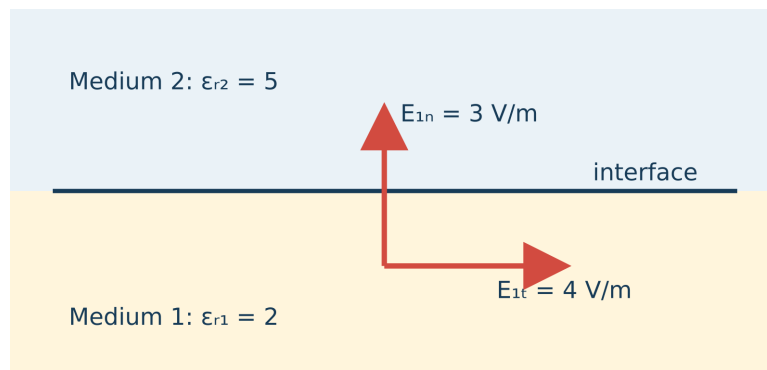


Figure: Charge-free dielectric interface

Accessible description: Horizontal dielectric interface with a normal axis, a tangential axis, and electric-field components in medium 1.

Q57. [1 mark; MCQ]

Audit ID: TMB-GATE-EE-EMFT-004 · revision 1 · Medium

Two concentric conducting spheres of radii a and b , with $b > a$, are separated by a dielectric of permittivity ϵ . Neglect fringing. Which expression gives the capacitance?

- **A.** $4\pi\epsilon(a + b)$
- **B.** $4\pi\epsilon \frac{ab}{b-a}$
- **C.** $4\pi\epsilon \frac{b-a}{ab}$
- **D.** $\frac{\epsilon ab}{4\pi(b-a)}$

Q58. [2 marks; NAT]

Audit ID: TMB-GATE-EE-EMFT-005 · revision 1 · Medium

A solid sphere of radius 0.1 m carries a uniform volume charge with total charge 8 nC in free space. Determine the potential at the centre relative to the potential at the surface, in volts. Use $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

NAT response: _____**Q59. [1 mark; NAT]**

Audit ID: TMB-GATE-EE-EMFT-006 · revision 1 · Easy

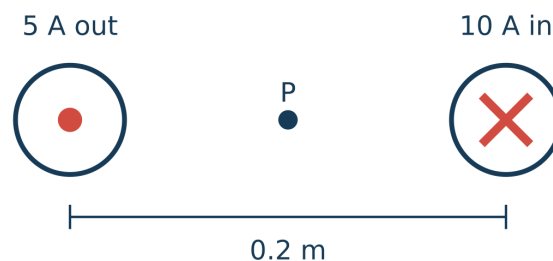
A parallel-plate capacitor has half of its plate area filled through the full separation with a dielectric of relative permittivity 3; the other half remains air. Neglect fringing. Enter the ratio of its capacitance to the all-air capacitance.

NAT response: _____**Q60. [2 marks; MCQ]**

Audit ID: TMB-GATE-EE-EMFT-007 · revision 1 · Medium

The figure shows two infinitely long parallel conductors separated by 0.2 m. The left conductor carries 5 A out of the page and the right conductor carries 10 A into the page. Determine the magnetic-flux-density magnitude at the midpoint.

- A. $10 \mu\text{T}$
- B. $15 \mu\text{T}$
- C. $30 \mu\text{T}$
- D. $45 \mu\text{T}$

Figure: Parallel conductors and midpoint P

Accessible description: Two conductors 0.2 m apart, with 5 A out of the page on the left and 10 A into the page on the right; midpoint P is marked.

Q61. [1 mark; MSQ]

Audit ID: TMB-GATE-EE-EMFT-008 · revision 1 · Medium

Select all statements that are valid in magnetostatics for ordinary media without magnetic monopoles.

- **A.** $\oint_C \mathbf{H} \cdot d\mathbf{l}$ equals the free current enclosed by C .
 - **B.** $\nabla \cdot \mathbf{B} = 0$.
 - **C.** A free surface-current density can produce a discontinuity in tangential $\mathbf{H}$.
 - **D.** The normal component of $\mathbf{B}$ may have an arbitrary discontinuity even when no magnetic monopoles exist.
-

Q62. [2 marks; NAT]

Audit ID: TMB-GATE-EE-EMFT-009 · revision 1 · Hard

A toroidal magnetic core has mean magnetic path length 0.5 m, relative permeability 1000, and a single air gap of length 1 mm. A 500-turn winding carries 0.2 A. Neglect leakage and fringing. Determine the flux density in tesla.

NAT response: _____**Q63. [1 mark; MCQ]**

Audit ID: TMB-GATE-EE-EMFT-010 · revision 1 · Medium

A single-turn loop links magnetic flux $\Phi(t) = (4t^2 + 2t)$ mWb. Determine the magnitude of the induced emf at $t = 0.5$ s.

- **A.** 2 mV
 - **B.** 4 mV
 - **C.** 6 mV
 - **D.** 8 mV
-

Q64. [2 marks; NAT]

Audit ID: TMB-GATE-EE-EMFT-011 · revision 1 · Medium

A conducting rod of length 0.5 m moves at 4 m/s perpendicular to a uniform magnetic field of 0.8 T. The rod completes a circuit of total resistance $2\ \Omega$. Neglect friction. Enter the external force required to maintain constant speed, in millinewtons.

NAT response: _____**Q65. [2 marks; MCQ]**

Audit ID: TMB-GATE-EE-EMFT-012 · revision 1 · Hard

The cross-sectional figure shows a long solenoid of radius 0.1 m whose uniform internal magnetic flux density changes at $dB/dt = 2$ T/s. Neglect the field outside the solenoid. Determine the induced electric-field magnitude on a circular path of radius 0.2 m concentric with the solenoid.

- **A.** 0.0125 V/m
- **B.** 0.025 V/m
- **C.** 0.05 V/m
- **D.** 0.10 V/m

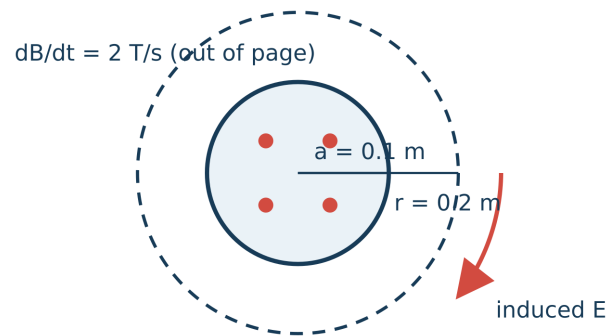


Figure: Solenoid cross-section and circular observation path

Accessible description: Concentric circles of radii 0.1 m and 0.2 m representing the solenoid cross-section and the integration path, with increasing axial magnetic field.

End of question paper

Release remains blocked until this exact question PDF and its paired solution PDF pass named-human complete-paper technical and visual QA.